

UNIVERSITAT POLITÈCNICA DE CATALUNYA

School of Professional & Executive Development

Postgrado en

Mobile Apps Design

BROADCAST RECEIVERS

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BROADCAST RECEIVERS

- The programming of broadcast receivers is a useful communications tool among components of the same app. or of different apps.
- It is also the communication tool for system events like: receiving a SMS, a battery level change, the phone switch on, Wi-Fi availability, new GPS coordinates, etc.

BROADCAST RECEIVERS

- We program a Java class as an extension of BroadcastReceiver and over-write the method **onReceive**():

```
public class MyBroadcastReceiver extends BroadcastReceiver {
@Override
    public void onReceive(Context context, Intent intent) {...}
}
```

BROADCAST RECEIVER

- We register a broadcast receiver for a given action type, for example, as follows:

```
myReceiver = new MyBroadcastReceiver();
IntentFilter myFilter = new IntentFilter("an_action");
registerReceiver(myReceiver, myFilter);
```

- And to unregister, as follows:

```
unregisterReceiver(myReceiver);
```

- **Note**: you can not unregister a non registered broadcast receiver, it throws an exception

BROADCAST RECEIVER

- We can create by code a **BroadcastReceiver** and an **IntentFilter** to register the BroadcastReceiver with the associated IntentFilter.
- On the other hand, we can declare a **BroadcastReceiver** on the **Manifest** file, including the **IntentFilter** for that BroadcastReceiver.

BROADCAST RECEIVER

- Android delivers an intent to all registered broadcast receivers when the event happens, and executes the **onReceive**() method of each of them.
- A component can send a broadcast message (equivalent to a system event) passing an intent object on calling:
  - **sendBroadcast**(intent), to all at the same time
  - **sendOrderedBroadcast**(intent), one at a time according to android:priority attribute at the IntentFilter
  - **sendStickyBroadcast**(…) / **removeStickyBroadcast**(…)

BROADCAST RECEIVER

- An easy form of communication, between a service (executing something in background) and an activity, comes from registering a broadcast receiver in the activity for a given broadcast **action** defined by the same activity.
- The service, after finishing its work, to asynchronously communicate its outcome, sends a broadcast message with the predefined **action**, and the attached Intent object will contain all the necessary Extras to deliver the task result.